Same Position, Two Look-Up Tables:

Novice (square → square) vs. Master (pattern → chunk) [Chase & Simon, 1973]

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.

Board flashed for 5 seconds

~25 pieces from a real game,

then reconstruct from memory

piece by piece

pattern chunks

Novice Player

Novice Look-Up Table
one row per single piece

Isolated Piece	Memory Slot
White pawn on e4	Slot 1
Black knight on f6	Slot 2
White bishop on c4	Slot 3
Black pawn on d6	Slot 4
... ~25 separate pieces	... slots run out (~7)

Outcome

Only ~6 pieces recalled;

working memory overflows

Chess Master

Master Look-Up Table
one row per recognized chunk

Recognized Pattern	Stored as One Chunk
Castled king + pawn shield (g1,f2,g2,h2)	"Kingside castle" = 1 chunk
Ruy Lopez opening structure	"Ruy Lopez" = 1 chunk
Isolated queen-pawn formation	"IQP" = 1 chunk
... a handful of familiar gestalts	... a few big chunks

Outcome

Nearly the whole board recalled

(edge vanishes on random boards)